

Effects of Powder Reuse in Metal Additive Manufacturing:
A Systematic Review

Scope: powder bed fusion (PBF-LB and PBF-EB); eight alloy families



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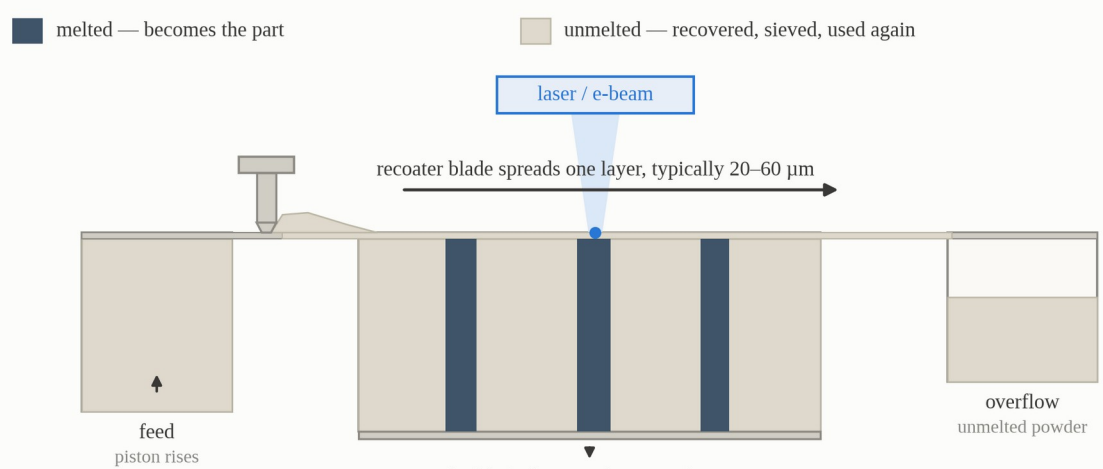
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1 · Introduction

Powder bed fusion melts only a small amount of the powder spread in each build. The unused powder is sieved and reused. Reuse decides the feedstock cost: one cycle cuts part cost by about a third (Ferreira 2025), and ignoring reuse can make a build look up to 75% cheaper than it is (Barclift 2016).

At every build the powder is exposed to heat, oxygen and spatter. It picks up oxygen, its particle size drifts with the sieving routine, and the particle surfaces change. Studies disagree on how far this can go before the part suffers.

This systematic review study shows the current evidence on powder reuse across alloy families and identifies critical gaps in the properties evaluated.



2 · Materials and Methods

PRISMA 2020 systematic review of powder reuse in powder bed fusion. Four searches in Scopus and Web of Science (27–30 August 2026; English, 2014 onward): one in each database, plus two more in Scopus for Co-Cr and other wording. Eligibility criteria were set before screening. Screening was LLM-assisted, and the first author decided every uncertain record. Recall was tested against 56 known papers: 51 came back (91%).

Every value in the text was checked against the source paper.

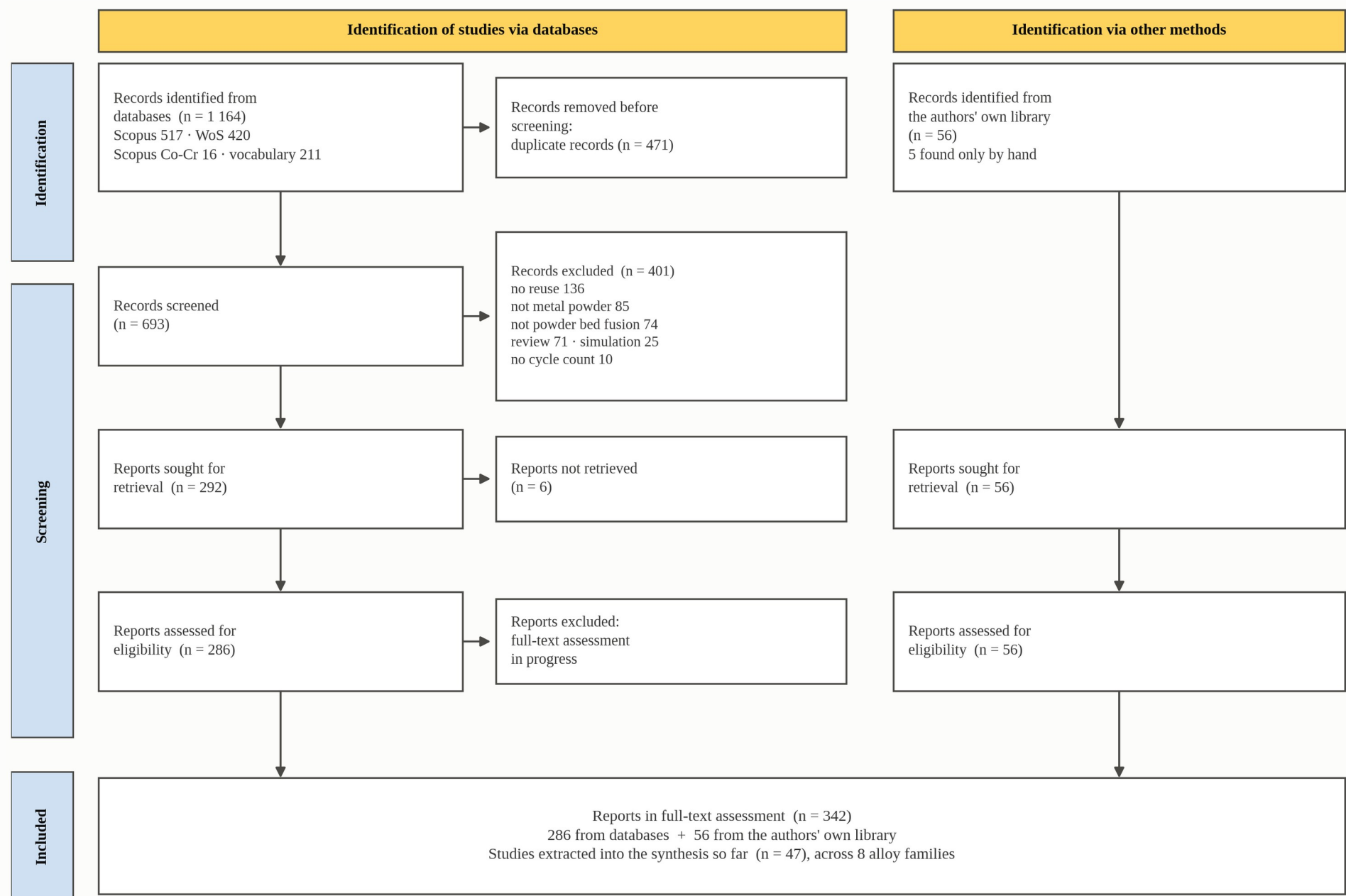


Figure 1. PRISMA 2020 flow. Layout after Page MJ et al., BMJ 2021;372:n71 (CC BY 4.0).

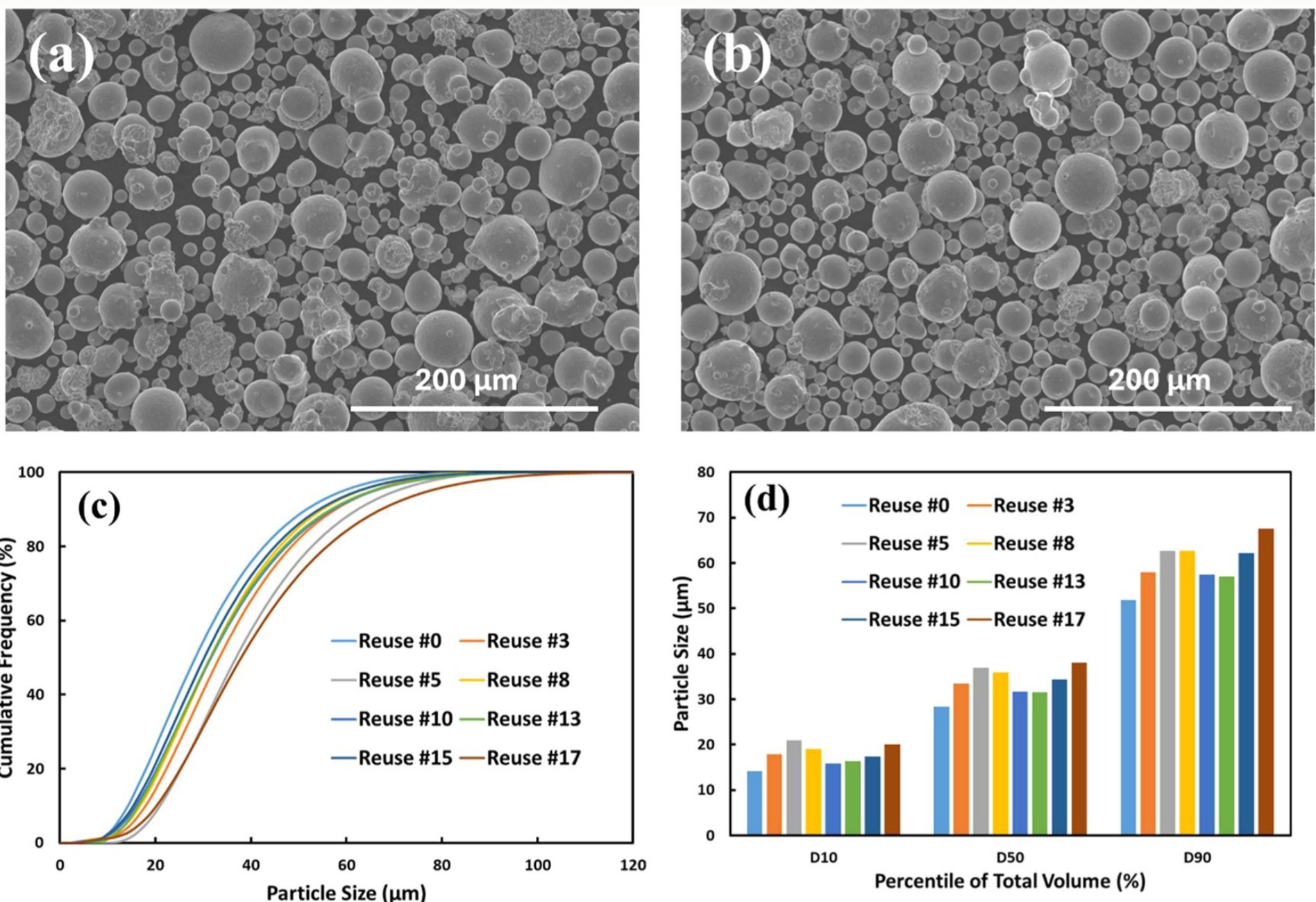


Figure 2. Co-Cr powder, Reuse #0 (a) and Reuse #17 (b); (c, d) size distributions; D50 rises 28.4 → 38.0 μm. Ghayoor et al., Addit. Manuf. Lett. 18 (2026) 100380, Fig. 3, CC BY-NC-ND 4.0.

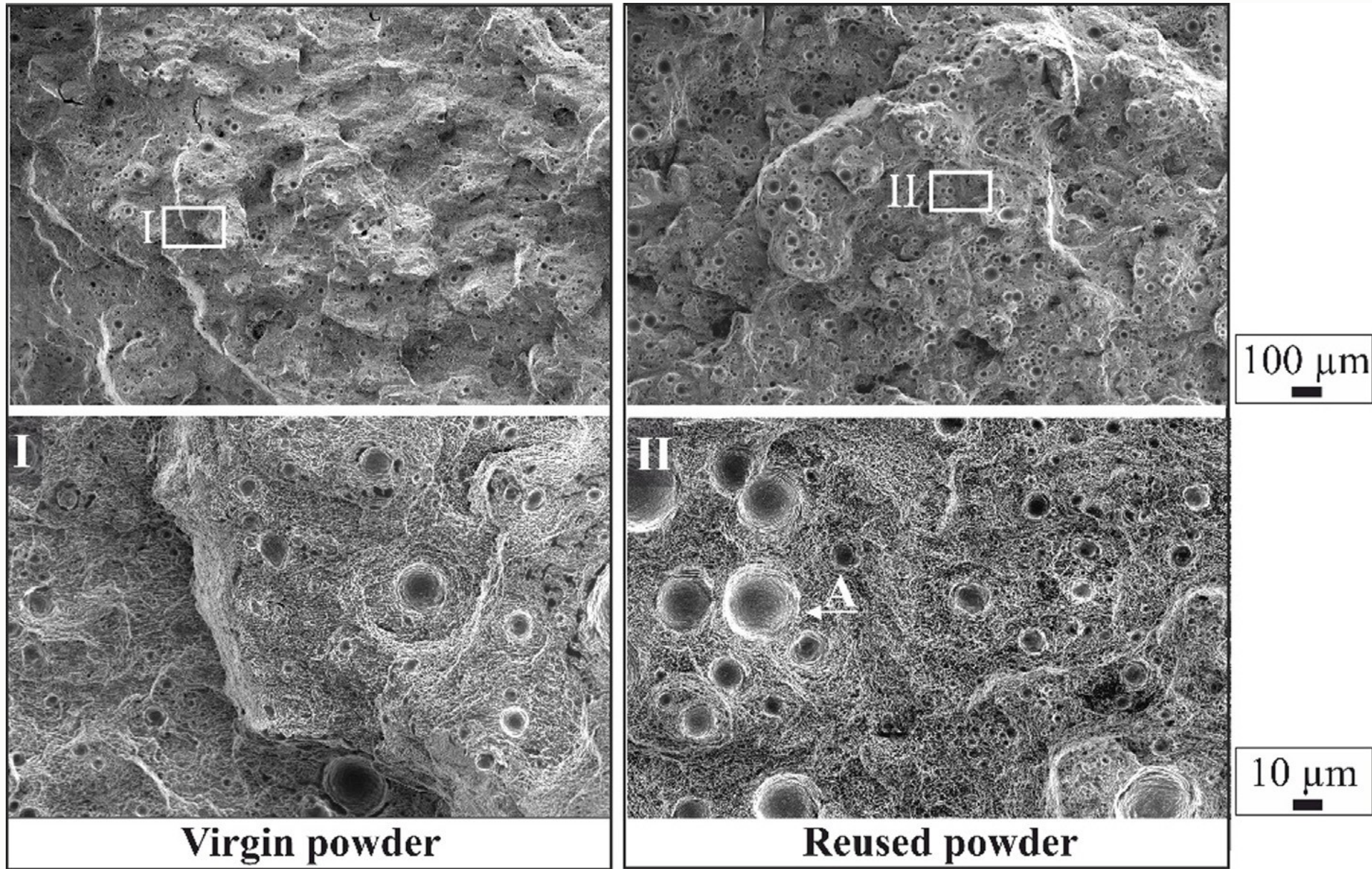


Figure 3. AISi10Mg fracture surfaces, virgin (left) and reused (right): more and larger gas pores. Fiegl et al., Mater. Des. 212 (2021) 110176, Fig. 5, CC BY-NC-ND 4.0.

4 · Discussion

Why the studies disagree. Tang and Schur ran the same alloy to similar oxygen levels with opposite outcomes (Figure 6): machines, atmospheres and sieving routines differ.

The handling sets the limit, not the powder. With virgin powder added and the atmosphere controlled, powder lasts much longer. Pure copper failed at cycle 8–9 with sieving alone, but ran to 59 cycles with top-up. Ti-6Al-4V reached about 100 builds (Figure 5).

What is missing. Impact energy, hardness, fatigue and flowability are the least reported properties (Figure 4). For Co-Cr none of the four has been measured, and neither has the powder oxygen content. This is the gap our project addresses.

3 · Results

So far 47 studies across eight alloy families have been extracted; 39 give a reuse number. In the matrix built here, 75 of 104 alloy-by-property cells have data. Impact energy for two, hardness and fatigue for four each.

What changes with reuse — and what has never been measured

POWDER					BUILT COMPONENT								
	Oxygen content	Particle size	Morphology, satellites	Flowability	Porosity in part	UTS	Yield strength	Elongation	Hardness	Fatigue (HCF)	Impact energy	Micro-structure	Phase / functional
Ti-6Al-4V	↑	↗	↓	↑	—	—	—	—	↗	↗	—	—	—
	0.08–0.19 wt%, 21 cyc	LB coarsens, EB narrows	↑ damaged particles rise	↑ improves in PBF-EB	—	+980–1000 MPa, 30 cyc	—	<60% over 30 builds	tracks O content	—	<30% vert., <12% horiz	—	—
21 papers	2.37	2.17	29.16	29.6	29.6	15.26	15.22	15.22,27	9.29	19.38	27	29.15	—
IN718	↑	—	—	—	—	—	—	—	—	—	—	—	↗
	146–266 ppm, 14 cyc	D50 rises, 14 cyc	—	—	—	stable w/ 50 cyc	—	—	HV 400–450 (add)	—	—	—	minor Layer plane
6 papers	238.08	39.1	8	3.18	8	23.1,13	108.13	23.1	39.1	33.23	—	39.1	18
Stainless steel	↑	—	—	—	—	—	—	—	—	—	—	—	—
	slow, within limits	—	—	—	—	—	—	—	—	↓ batch 15 vs batch 1	↓ declines from cyc. 1	—	↗ slight &-ferrite
5 papers	2.3	3.4	3	4.3	3.20	3.20,11	3.31	3.20	4	—	—	20	3.20
Co-Cr / CoCrMo	—	↑	↗	↗	—	—	—	↗	↗	—	—	—	↑
	—	13 and 17 cycles	↑ seen, not quantified	—	—	—	—	no systematic trend	—	—	—	—	Co, Cr ions (add, R2–R10)
5 papers	—	30.34	—	—	34.26	39.34	34	—	36.26	—	—	36.30	15
Al alloys	↑	↗	↗	—	—	↗	↗	↗	↗	↗	—	—	—
	fastest O pickup	minor, with sieving	oxide layer thickens	—	nil-at threshold	>17% at threshold	follows porosity	follows porosity	108–96 HV0.5	160–145 MPa	—	—	—
6 papers	14.21	14	21.14	24.14	12.57	5.12,24	5	5	12	5	—	24	—
Pure Cu	↗	—	—	—	—	—	—	—	—	—	—	—	—
	<170 ppm, 30 cyc	—	—	<2.85%	>99.38% dense	—	—	—	—	—	—	—	>99.5% IACS
1 paper	31	31	31	31	31	—	—	—	—	—	—	31	31
GH3536	↑	↗	↗	↗	—	↗	↗	↗	↗	—	—	—	—
	surface O lines	fewer fines, 8–16	satellites, rounds	—	—	slight decrease	slight decrease	slight increase	—	—	—	—	—
1 paper	31	31	31	—	—	31	31	31	31	—	—	31	—
NiTi	—	—	↑	—	—	—	—	—	—	—	—	↑	—
	from cycle 1	—	from cycle 1	—	—	—	—	—	—	—	—	Al, Mo still up	—
1 paper	32	—	32	—	—	—	—	—	—	—	—	32	—
↑ rises ↗ mixed / route-dependent — unchanged ↗ never reported ↓ flagged as critical in the matrix													

Figure 4. How each property changes with reuse, by alloy family. Hatched cells have never been reported. Small numbers point to the papers below.

Every study with a reuse count — one point per study

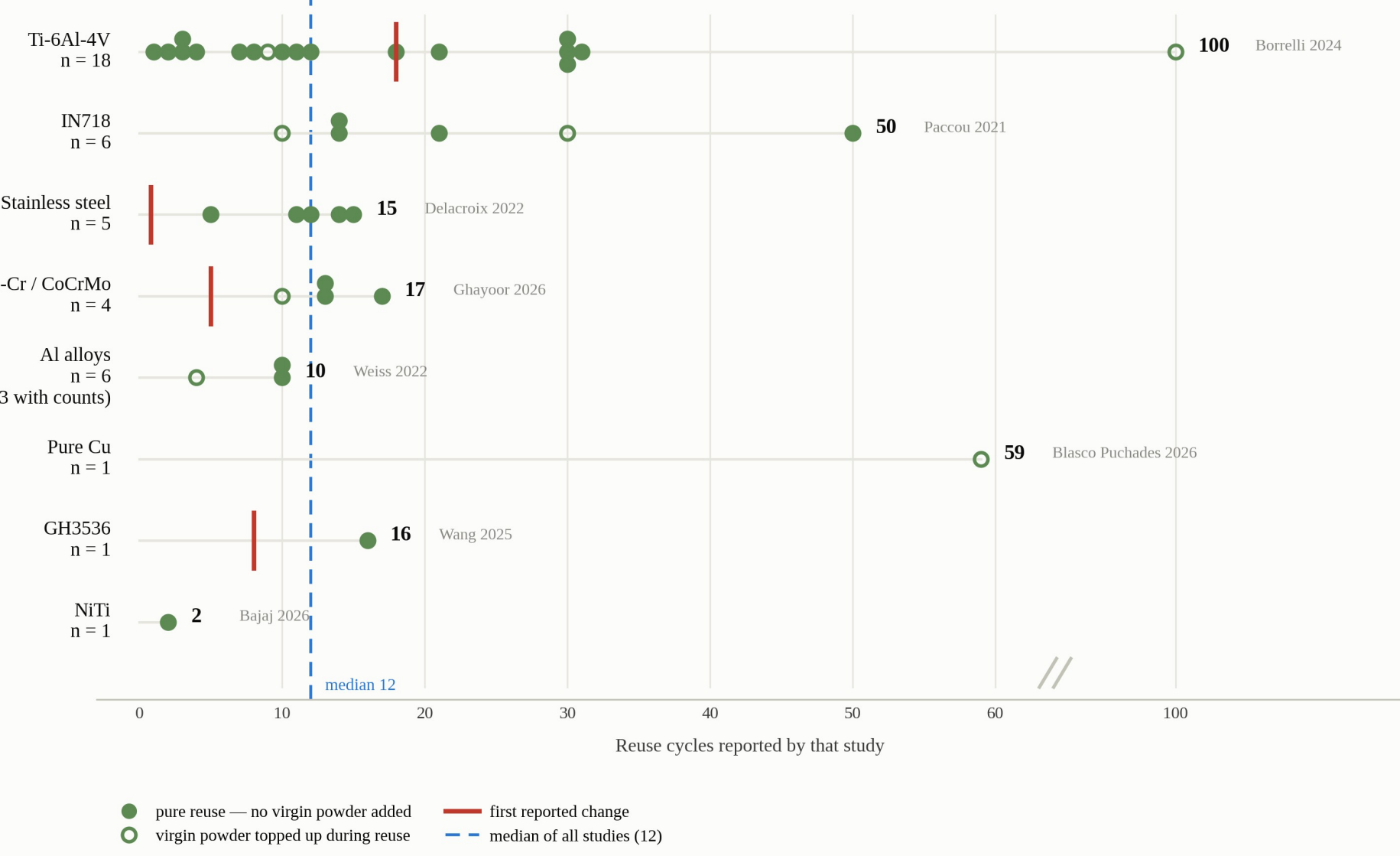


Figure 5. Every study that reports a reuse number, one point each; hollow points added virgin powder. Red line: the first reported change.

Same alloy, three protocols: oxygen pickup in Ti-6Al-4V

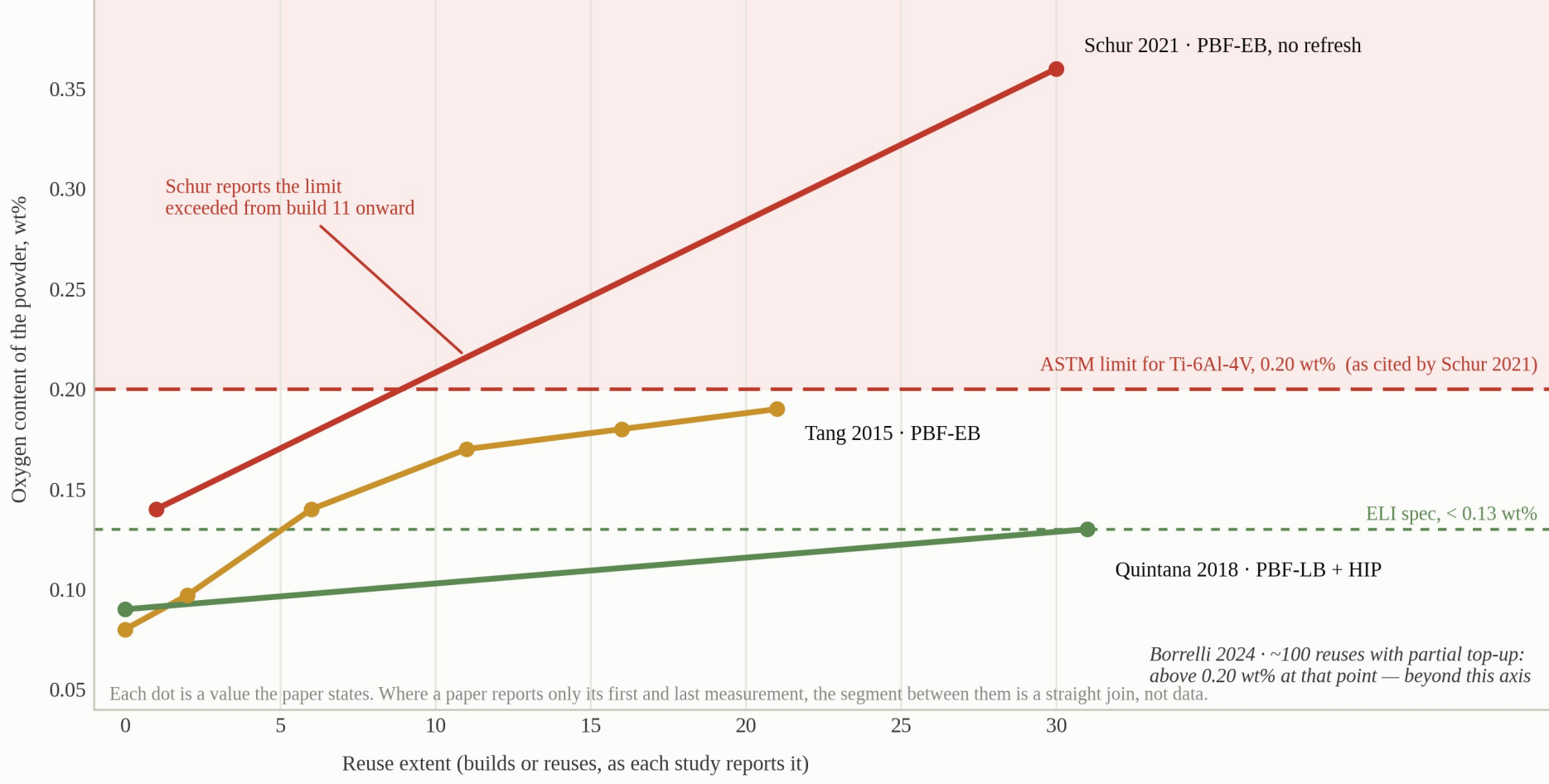


Figure 6. Ti-6Al-4V oxygen under three reuse protocols: the same alloy crosses the ASTM limit at build 11, or never.

Implications for practice

Watch oxygen, elongation and impact energy, not tensile strength. Tensile strength held or rose in every family where it was measured, except AISi10Mg and GH3536.

Judge the handling route, not the cycle count. One study crossed the 0.20 wt% oxygen limit at build 11. Another was still at 0.13 wt% after 31.

Dental Co-Cr: five reuses, 50–70% virgin added. Ion release rises while strength and density hold.

5 · Conclusion

Where powder chemistry was measured, it changed with reuse in every alloy, but the parts did not follow. Tensile properties held, and in Ti-6Al-4V they rose. Only AISi10Mg (–15%) and GH3536 lost strength. The losses showed up in elongation, impact energy and functional properties. Reported reuse spans 1 to about 100, median 12. The longest run ended above the ASTM oxygen limit.